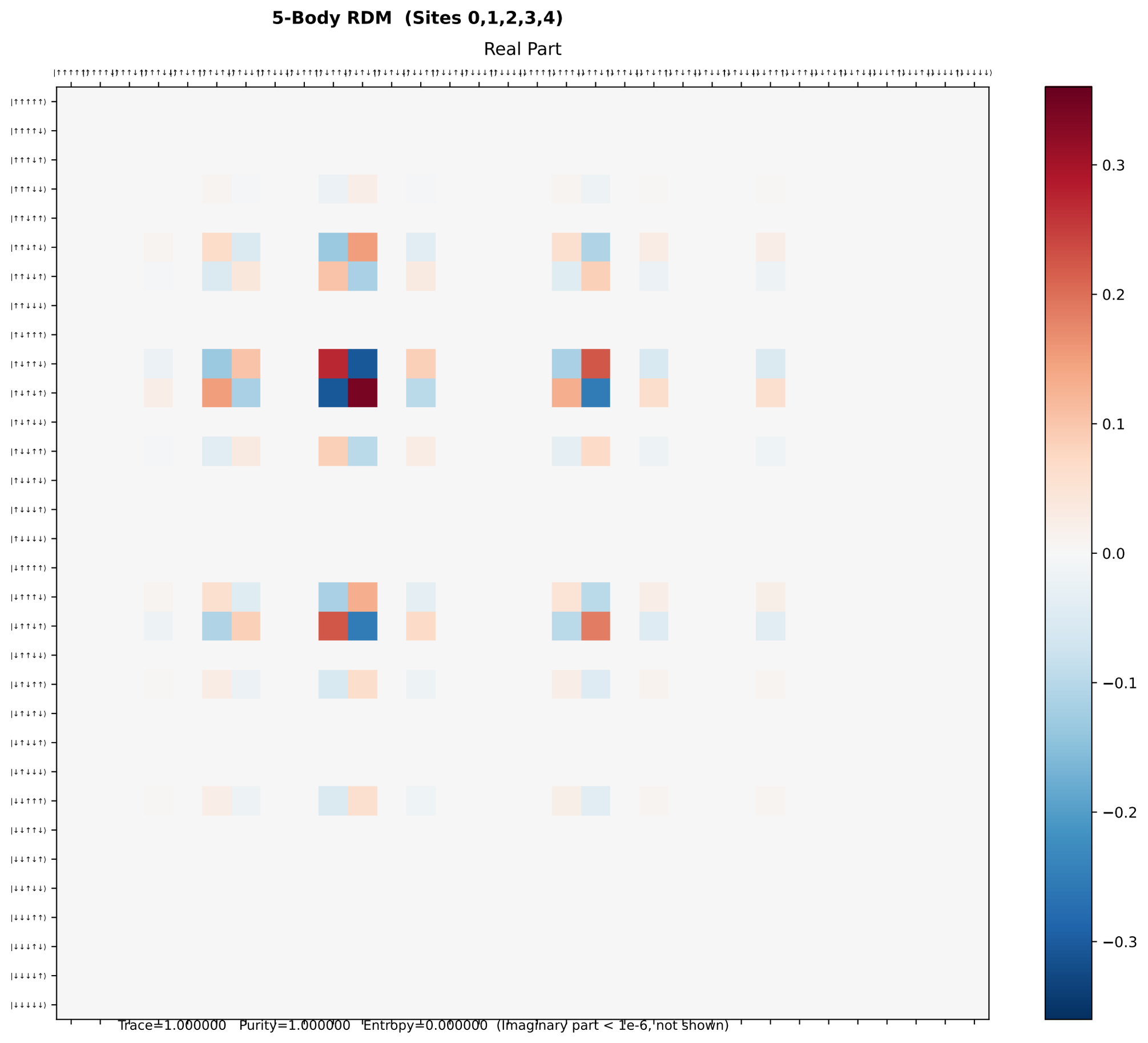
Genuine Multipartite Entanglement Analysis

```
Model                              Ground-state properties          GME summary
-----                              ----------------------           -----------
Hubbard ring (1-D, PBC)            E0                : -2.60008132  Extracted n-spin RDM : n = 5
                                   E1                : -2.60008132
System parameters                                                   Purity:
-----------------                  <S2>              : 0.750000       gamma = Tr(rho^2)
Sites (L)          : 5             S                 : 0.5000
Electrons (Ne)     : 5             <Sz>              : 0.250000     Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1 : 2.0            gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 4.0           Spin state        : Doublet (S=1/2)
Mode               : uniform                                       C_GME (Troiani et al.):
n-body order       : 5             Double occupancy                   C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
                                   ----------------
                                   <D> total         : 0.459102     Convex roof (mixed):
                                   <D> / site        : 0.091820       C_GME(ρ) = min Σ pᵢ C_GME(|ψᵢ))


                                                                    Date: 2026-04-10  20:59
```


5-Body RDM (Sites 0,1,2,3,4)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 0,1,2,3,4)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.681571e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
==================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
  (0,)         | (1, 2, 3, 4) | 0.62109914
  (0, 1)       | (2, 3, 4)    | 0.77570535
  (0, 2)       | (1, 3, 4)    | 0.48594088
  (0, 3)       | (1, 2, 4)    | 0.32113729
  (0, 4)       | (1, 2, 3)    | 0.32832759
  (0, 1, 2)    | (3, 4)       | 0.87682669
  (0, 1, 3)    | (2, 4)       | 0.42085618
  (0, 1, 4)    | (2, 3)       | 0.43357804
  (0, 2, 3)    | (1, 4)       | 0.28510065
  (0, 2, 4)    | (1, 3)       | 0.27956910
  (0, 3, 4)    | (1, 2)       | 0.57073601
  (0, 1, 2, 3) | (4,)         | 0.52552231
  (0, 1, 2, 4) | (3,)         | 0.50899047
  (0, 1, 3, 4) | (2,)         | 0.75535749
  (0, 2, 3, 4) | (1,)         | 0.54458614

STATUS: GME EXISTS
----------------------------------

C_GME = 0.49633318
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
==================================

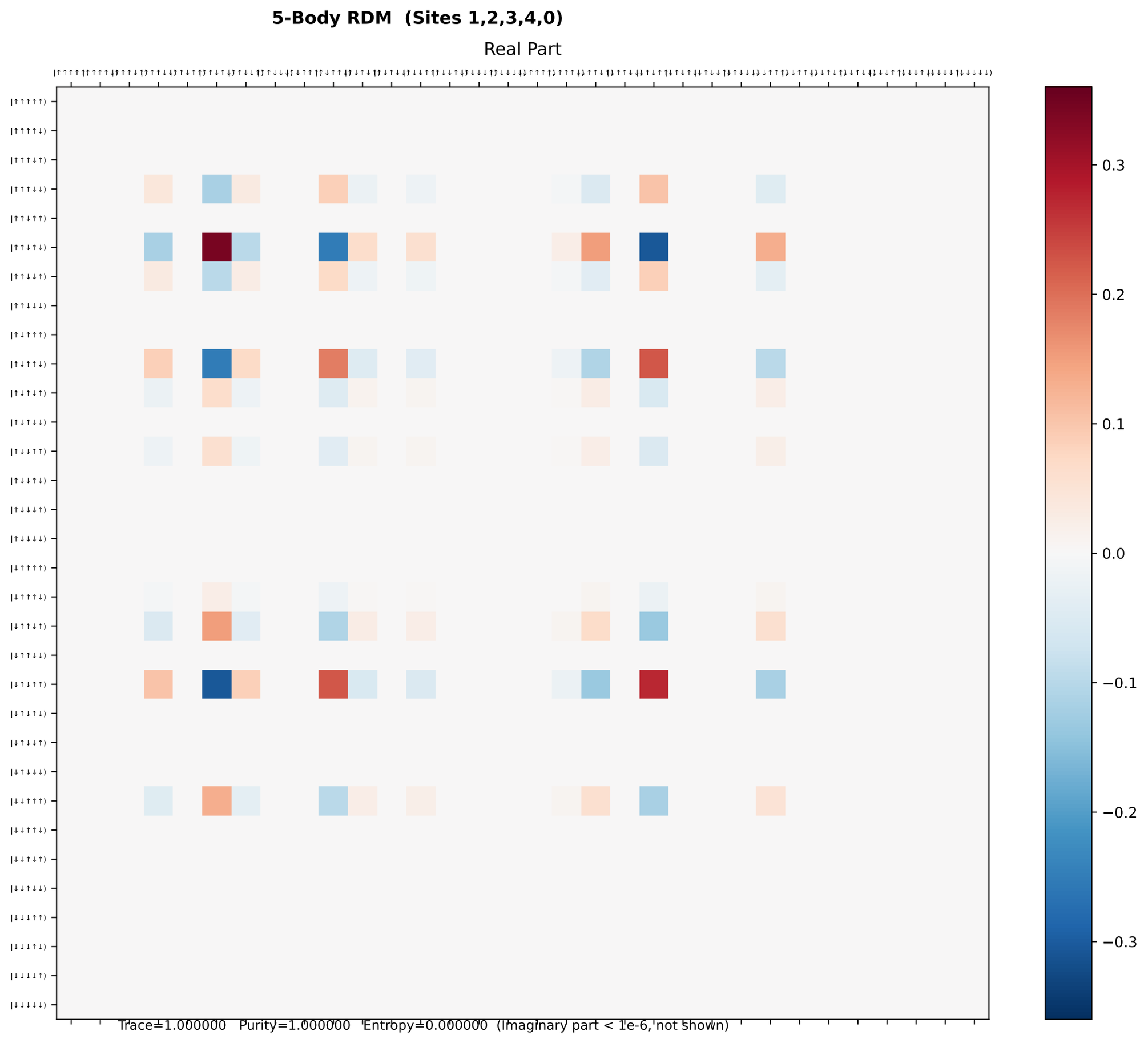
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


5-Body RDM (Sites 1,2,3,4,0)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 1,2,3,4,0)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-3.613134e-17
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 4, 0)	0.52552231
(1, 2)	(3, 4, 0)	0.32832759
(1, 3)	(2, 4, 0)	0.28510065
(1, 4)	(2, 3, 0)	0.42085618
(1, 0)	(2, 3, 4)	0.87682669
(1, 2, 3)	(4, 0)	0.43357804
(1, 2, 4)	(3, 0)	0.27956910
(1, 2, 0)	(3, 4)	0.57073601
(1, 3, 4)	(2, 0)	0.32113729
(1, 3, 0)	(2, 4)	0.48594088
(1, 4, 0)	(2, 3)	0.77570535
(1, 2, 3, 4)	(0,)	0.50899047
(1, 2, 3, 0)	(4,)	0.75535749
(1, 2, 4, 0)	(3,)	0.54458614
(1, 3, 4, 0)	(2,)	0.62109914

STATUS: GME EXISTS

C_GME = 0.49633318
GME confirmed across ALL cuts.

Convex Roof Analysis
==========

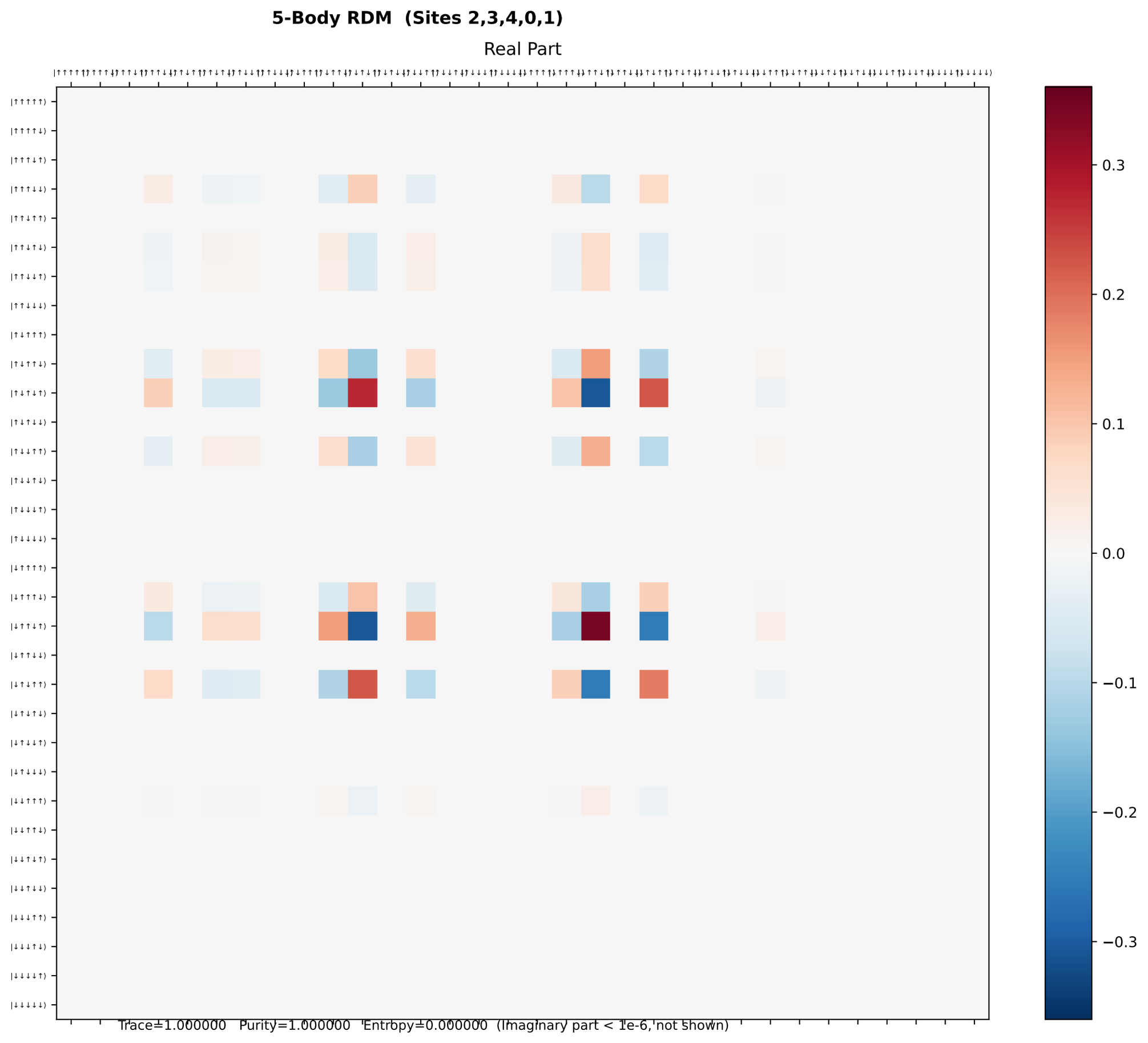
Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.00000000
- Best C_LB = 0.00000000

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.49633318 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.49633318 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.49633318

Status: GME
Bounds consistent: YES
Convergence: TIGHT


5-Body RDM (Sites 2,3,4,0,1)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 2,3,4,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -3.590841e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
  (2,)         | (3, 4, 0, 1) | 0.50899047
  (2, 3)       | (4, 0, 1)    | 0.87682669
  (2, 4)       | (3, 0, 1)    | 0.32113729
  (2, 0)       | (3, 4, 1)    | 0.27956910
  (2, 1)       | (3, 4, 0)    | 0.43357804
  (2, 3, 4)    | (0, 1)       | 0.57073601
  (2, 3, 0)    | (4, 1)       | 0.48594088
  (2, 3, 1)    | (4, 0)       | 0.77570535
  (2, 4, 0)    | (3, 1)       | 0.42085618
  (2, 4, 1)    | (3, 0)       | 0.28510065
  (2, 0, 1)    | (3, 4)       | 0.32832759
  (2, 3, 4, 0) | (1,)         | 0.75535749
  (2, 3, 4, 1) | (0,)         | 0.54458614
  (2, 3, 0, 1) | (4,)         | 0.62109914
  (2, 4, 0, 1) | (3,)         | 0.52552231

STATUS: GME EXISTS
------------------------------------

C_GME = 0.49633318
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
====================================

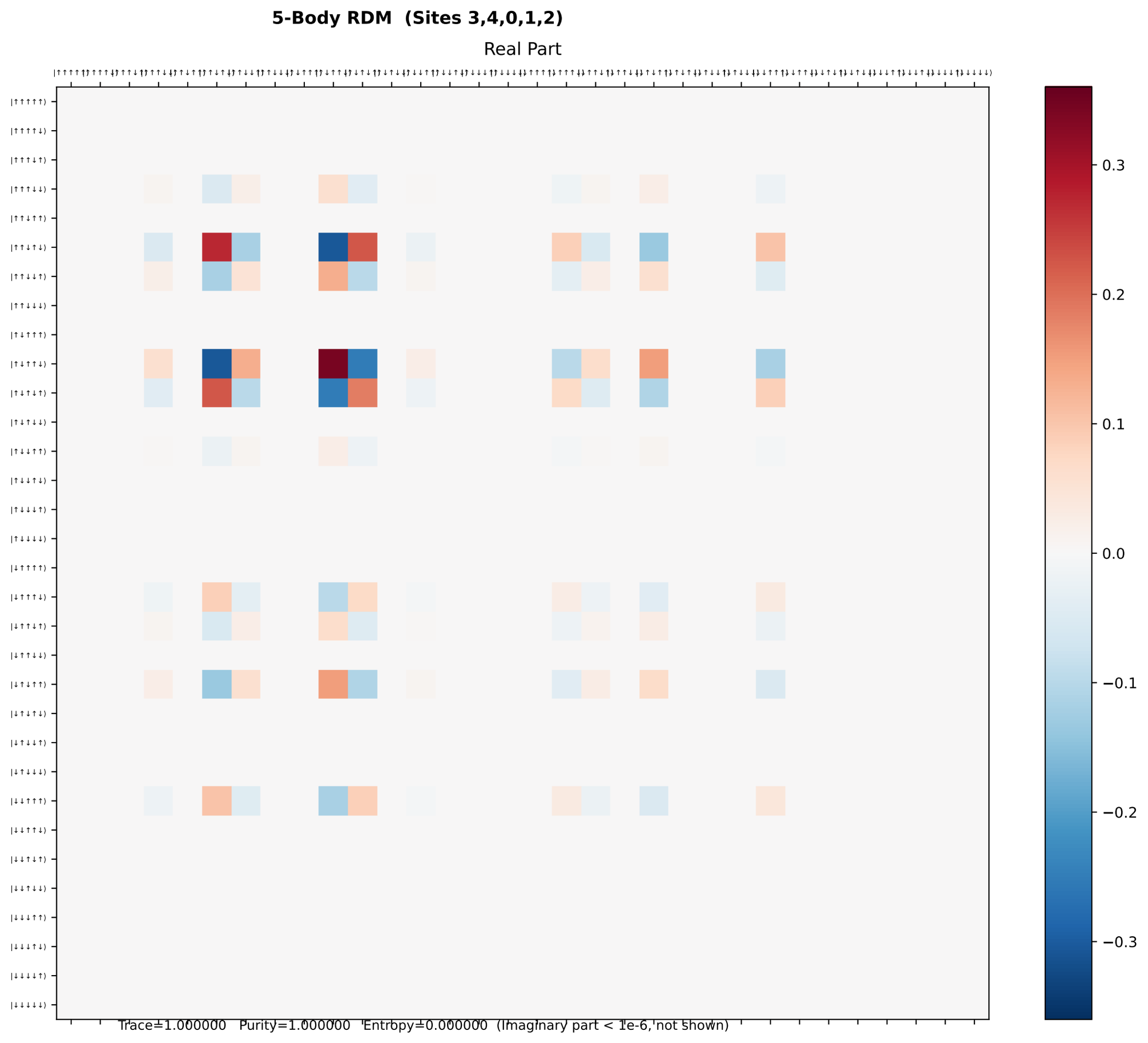
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


5-Body RDM (Sites 3,4,0,1,2)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 3,4,0,1,2)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -8.670812e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
  (3,)         | (4, 0, 1, 2) | 0.75535749
  (3, 4)       | (0, 1, 2)    | 0.43357804
  (3, 0)       | (4, 1, 2)    | 0.42085618
  (3, 1)       | (4, 0, 2)    | 0.48594088
  (3, 2)       | (4, 0, 1)    | 0.57073601
  (3, 4, 0)    | (1, 2)       | 0.77570535
  (3, 4, 1)    | (0, 2)       | 0.28510065
  (3, 4, 2)    | (0, 1)       | 0.32832759
  (3, 0, 1)    | (4, 2)       | 0.27956910
  (3, 0, 2)    | (4, 1)       | 0.32113729
  (3, 1, 2)    | (4, 0)       | 0.87682669
  (3, 4, 0, 1) | (2,)         | 0.54458614
  (3, 4, 0, 2) | (1,)         | 0.62109914
  (3, 4, 1, 2) | (0,)         | 0.52552231
  (3, 0, 1, 2) | (4,)         | 0.50899047

STATUS: GME EXISTS
------------------------------------

C_GME = 0.49633318
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
====================================

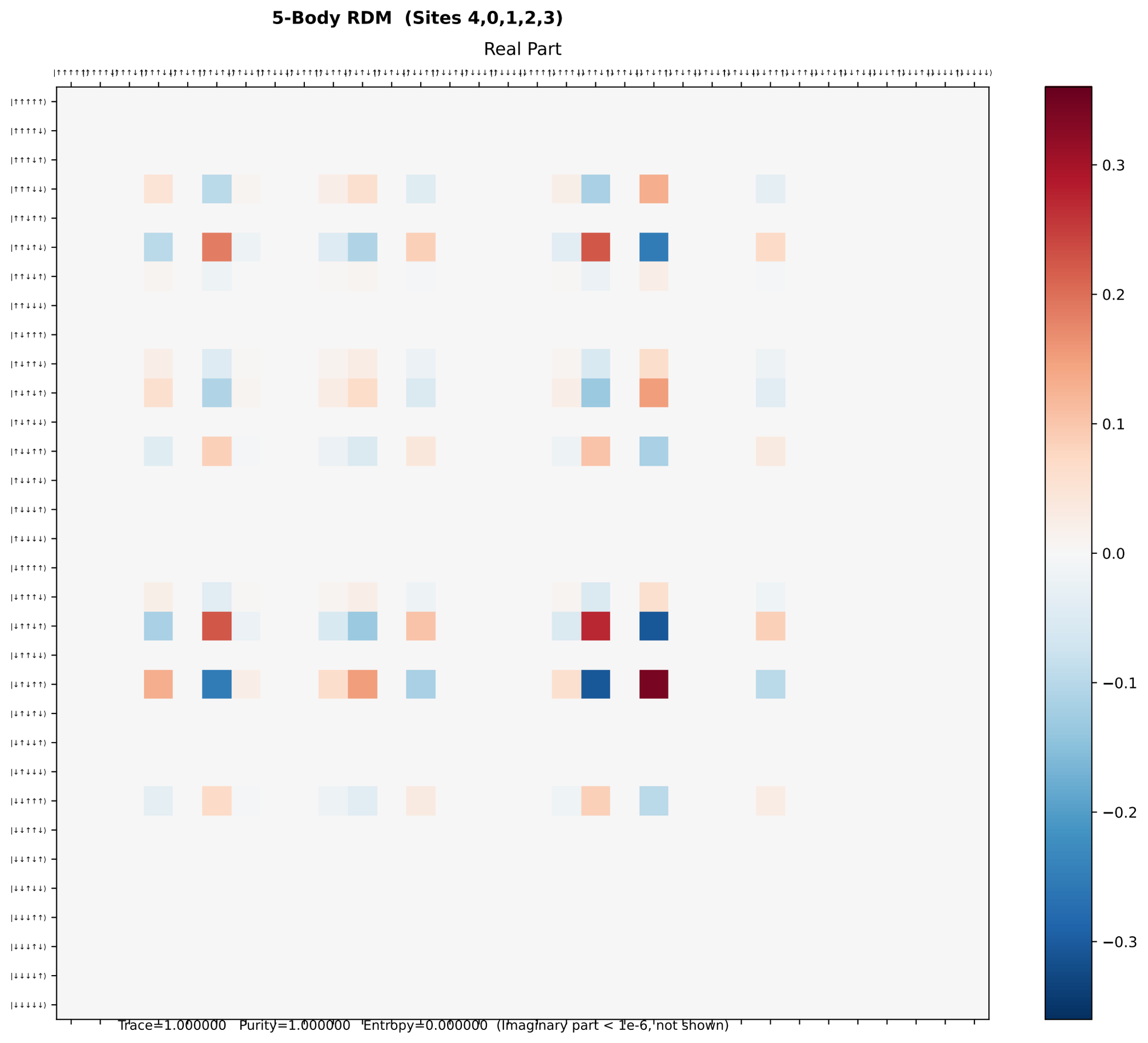
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


5-Body RDM (Sites 4,0,1,2,3)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 4,0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.249933e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
  (4,)         | (0, 1, 2, 3) | 0.54458614
  (4, 0)       | (1, 2, 3)    | 0.57073601
  (4, 1)       | (0, 2, 3)    | 0.27956910
  (4, 2)       | (0, 1, 3)    | 0.28510065
  (4, 3)       | (0, 1, 2)    | 0.77570535
  (4, 0, 1)    | (2, 3)       | 0.32832759
  (4, 0, 2)    | (1, 3)       | 0.32113729
  (4, 0, 3)    | (1, 2)       | 0.87682669
  (4, 1, 2)    | (0, 3)       | 0.48594088
  (4, 1, 3)    | (0, 2)       | 0.42085618
  (4, 2, 3)    | (0, 1)       | 0.43357804
  (4, 0, 1, 2) | (3,)         | 0.62109914
  (4, 0, 1, 3) | (2,)         | 0.52552231
  (4, 0, 2, 3) | (1,)         | 0.50899047
  (4, 1, 2, 3) | (0,)         | 0.75535749

STATUS: GME EXISTS
------------------------------------

C_GME = 0.49633318
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```